

PC Build Store

Individual Contribution Details

Object Oriented Programming - Final Submission

2025-2026

Abdulhanan

rehanazizabdulhanan@gmail.com

Role
Team Lead / Full Stack

GUI Contributions

- DashboardGUI - Hero banner, stats cards, featured parts, navigation
- ReportGUI - Revenue line chart, pie chart, bar chart, custom painting
- ChatSettingsDialog - Modal settings form for AI configuration
- AddPartDialog - Modal form for adding components via UI
- Theme.java - Dark theme color palette and styling constants
- Components.java - DarkScrollBarUI, button factories, UI helpers

Backend Contributions

- ChatService - SSE streaming via java.net.http.HttpClient
- ChatConfig - Settings persistence in ~/.pcbuildstore/config.json
- ReportDAO - Revenue aggregation, parts distribution, performance queries
- ImageCache - 3-tier caching (memory, disk, HTTP) with 8-thread parallel
- Main.java - Application entry point

Database Work

- Database schema design (all 6 tables)
- Seed data: 78+ components with real Pakistan market pricing
- Migration scripts for image paths and bulk updates

Testing & Debugging

- Chat streaming with KoboldCPP (Qwen3 model)
- Dashboard stats accuracy
- Report chart data validation
- Image cache hit/miss rates

Ahmad Mushtaq

2501803@students.au.edu.pk

Role

Billing Module Lead

GUI Contributions

- BillingGUI - Build summary display, purchase confirmation
- Receipt view with formatted HTML output
- Print receipt via `java.awt.print.PrinterJob`
- Save receipt as HTML file via `JFileChooser`

Backend Contributions

- BillDAO - Purchase processing, bill creation, retrieval
- Bill model - Purchase record POJO
- HTML receipt generation with table layout
- Printable interface implementation

Database Work

- Bill insertion queries
- Bill retrieval with JOIN to builds
- Purchase date formatting

Testing & Debugging

- Purchase flow end-to-end
- Receipt print dialog verification
- Save receipt file output validation
- Database insert confirmation

Aitzaz

2501805@students.au.edu.pk

Role

Build Upgrades Module Lead

GUI Contributions

- BuildUpgradesGUI - 6-slot upgrade interface
- Current parts display with specs
- Upgrade suggestions panel
- Apply upgrades button

Backend Contributions

- GPUOptionDAO - GPU-specific upgrade queries
- GPUOption model - GPU upgrade option POJO
- Compatibility checking algorithms
- Socket compatibility matrix (Intel/AMD)
- DDR generation validation
- Wattage headroom calculation

Database Work

- Build part replacement queries
- Build score recalculation
- GPU options retrieval

Testing & Debugging

- Compatibility check accuracy
- Socket mismatch detection
- DDR generation validation
- Wattage overflow detection

Haroon

haroonikram2512@gmail.com

Role

Build Configurator Lead

GUI Contributions

- BuildCatalogGUI - Category tabs, product grid
- Cart sidebar with selected parts
- 6-slot build panel with motherboard
- Product cards with images and specs

Backend Contributions

- BuildDAO - Build CRUD, part management
- Build model - PC configuration POJO
- BuildPart model - Junction table POJO
- CategoryDAO - Category retrieval
- PartDAO - Component queries and filtering

Database Work

- Build insertion and retrieval
- Build parts management
- Part filtering by category
- Category listing queries

Testing & Debugging

- Category tab switching
- 6-slot selection enforcement
- Price/score calculation
- Cart state persistence